MIL-R-9409B(USAF)
28 August 1962
Superseding
MIL-R-9409A(USAF)
16 June 1955

MILITARY SPECIFICATION

RACK, BOMB, AIRCRAFT, TYPE MA-4A

1. SCOPE

1.1 This specification establishes the requirements for one type of bomb rack, designated Type MA-4A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-W-6858	Welding; Low Carbon Steel, Austenitic Steels, and Nickel Alloys, Spot and Seam
MIL-S-6872	Soldering Process, General Specification for
MIL-N-7513	Nomenclature Assignment and Nameplate Approval, Contractors Method for Obtaining
MIL-L-7870	Lubricating Oil (General Purpose Low Temperature)
MIL-E-17555	Electronic Equipment and Associated Maintenance Parts; Preservation, Packaging, Packing and Marking of

STANDARDS

Military

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MIL-STD-143	Specifications and Standards; Order of Precedence for Selection of
MS3102	Connectors - Electrical, Receptacles, Box Mounting
MS3106	Connectors - Electrical, Plugs, Straight

DRAWINGS

U. S. Air Force

55J4312	Rack, Bomb, Aircraft, Type MA-4A
53A5455	Bushing, Adapter
53B5454	Bushing, Hook Pivot
60C98799	Hook, Front
60C98800	Hook, Rear

3. REQUIREMENTS

3.1 Preproduction sample. This specification requires the submission of preproduction sample for preproduction testing. See Section 4 for details.

3.2 Selection of specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.2.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.3 Design:

3.3.1 General. The type MA-4A rack shall be a 2-hook rack for internal or external installation in bomber or fighter-type aircraft. A separate emergency release lever shall be included also in the design to release the rack independent of electrical power.

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3.3.2 The bomb rack shall be designed in order that the configuration and assembly shall conform with Drawing No. 55J4312 and the details thereof.

3.3.3 Electrical characteristics. The bomb rack shall be designed to operate electrically on 28V dc.

3.3.3.1 Voltage. The rack's release circuit shall operate satisfactorily on 18 to 28 volts dc.

3.3.3.2 Current drain. The current drains of the solenoids at 28V dc and temperature 68°F (20°C) shall not exceed the following:

a. Release solenoids: 6 amperes

b. Arming solenoids: 0.5 ampere (The arming mechanism shall function satisfactorily at 18V dc.)

3.3.4 Time of release. Not more than .040 seconds shall elapse from beginning of electrical impulse through the release of circuit "B" Figure 1 to the time when the bomb carrying hooks begin to open.

3.3.5 Time of transfer. During operations of the release circuit "B" in Figure 1, the time from end of electrical impulse to completion of switching into the next rack circuit shall be a maximum of 17 milliseconds.

3.3.6 Separate release circuit. The rack is arranged so that a separate release circuit (jettison) can release the rack independent of the normal release circuit. This circuit shall be such that the impulse required to release the rack shall be transferred to an output wire (jettison output) immediately upon the opening of the hooks. With this circuit, it is then possible to release several racks in series with one impulse.

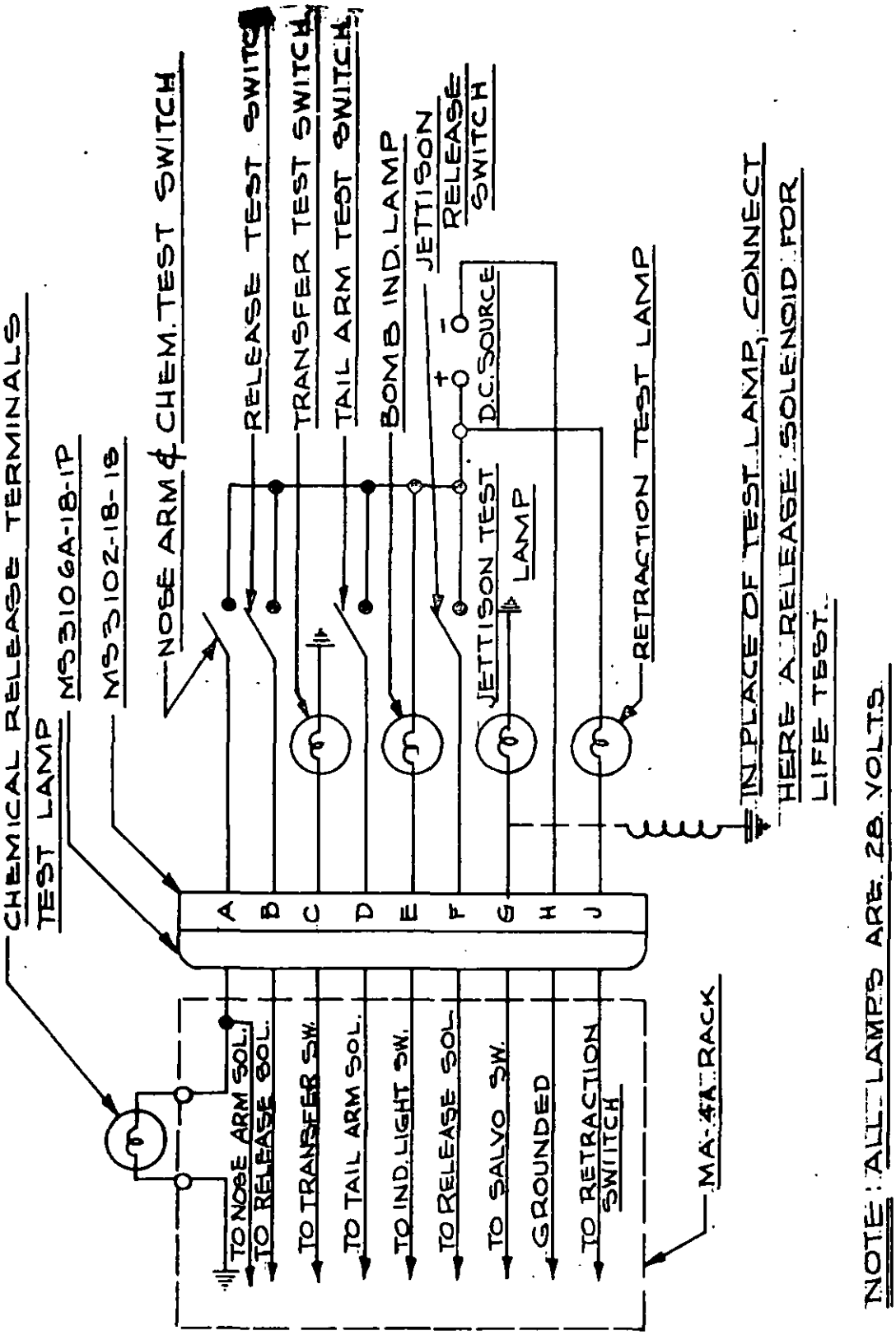
3.3.7 Temperature. The rack shall be designed for normal operation throughout the temperature range of -65°F to +160°F.

3.4 Construction. The Type MA-4A bomb rack shall be constructed so that no parts will work loose in service. The rack shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, storage, installation and service.

3.4.1 Springs. In order to prevent brittleness as a result of plating, all springs shall be treated in accordance with Specification QQ-P-416.

3.4.2 Welding. Any welding required in the construction of the rack shall be in accordance with Specification MIL-W-6858.

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TEST CIRCUIT

FIGURE I

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3.4.3 Soldering. Soldering required in the construction of the rack shall be in accordance with Specification MIL-S-6872.

3.4.4 Shock. The rack release shall be constructed so that shock will not cause tripping of the mechanism.

3.4.5 Lubrication. The rack shall operate without lubrication. If an initial lubrication is necessary in the sealed units only, it shall be limited to a slight wiping of the bearing surfaces with oil conforming to Specification MIL-L-7870.

3.4.6 Strength data. The following strength data shall apply with the Type MA-4A bomb rack shear-mounted in a rigid supporting structure.

	Load Condition	Ult. Load
Vertical Down:	2-3/8-inch High Shear Mounting Bolts	38,000 lbs.
Vertical Down:	2-5/16-inch High Shear Mounting Bolts using the Adapter Bushing, Drawing No. 53A5455	28,000 lbs.
Vertical Down:	2-5/16-inch AN Standard Mounting Bolts using the Adapter Bushing, Drawing No. 53A5455	23,000 lbs.
Forward:	5/16-inch or 3/8-inch AN Standard or High Shear Mounting Bolts	20,000 lbs.
Aft:	5/16-inch or 3/8-inch AN Standard or High Shear Mounting Bolts	10,000 lbs.

3.4.7 The bushing, adapter, bomb rack (Drawing No. 53A5455) shall be secured to the rack, one inside each hook bushing. (Drawing No. 53B5454). Method of securing shall be such that the bushing is not inadvertantly separated from the rack. Each adapter bushing shall be tagged, such tag containing, but not limited to, the following information:

a. "Remove this loose adapter bushing if 3/8-inch mounting bolts are used for installation.

b. "Leave bushing in place (inside hook bushing) if 5/16 inch mounting bolts are used."

3.5 Performance. The bomb rack shall be capable of operating under the conditions specified herein and shall meet the requirements of Section 4.

3.6 Marking, identification and nomenclature. Nomenclature approval and marking of parts and equipment for identification purposes shall be in accordance with Specification MIL-N-7513.

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3.7 Workmanship. The bomb rack, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. The inspection and testing of the Type MA-4A bomb rack shall be classified as follows:

a. Preproduction tests. Preproduction tests are those tests accomplished on samples which are representative of the production of the item after the award of the contract, to determine that the production meets the requirements of this specification.

b. Acceptance tests. Acceptance tests are those tests accomplished on the Type MA-4A bomb rack manufactured and submitted for acceptance under contract.

4.2 Test conditions. Unless otherwise specified, tests shall be conducted at temperature 68°F (20°C) and at a relative humidity of 25 to 50 per cent.

4.3 Preproduction tests:

4.3.1 Preproduction test samples. One rack, as specified in the contract, shall be subjected to the preproduction tests specified herein to determine compliance with this specification.

4.3.2 The preproduction tests shall consist of all the acceptance tests (4.4) and the following:

4.3.2.1 Vibration test. The rack shall be subjected to the vibration test specified in Procedure I of Specification MIL-E-5272.

4.4 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling tests.

4.4.1 Individual tests:

4.4.1.1 Examination of product. Each rack shall be checked to determine conformance to this specification with respect to material, workmanship, dimensions, and markings.

4.4.1.2 Arming mechanism. When the rack is cocked, the arming wire retainers must allow the arming wire swivel loops to be

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inserted and must retain them until the rack is released; however, the arming wires must be retained by the arming device when the rack is released after the arming circuits are closed (Figure 1). The former is known as a "safe" release and the latter is known as an "armed" release.

4.4.1.3 Electrical circuits. One terminal of each solenoid is grounded to the solenoid case, which is in turn grounded to the bomb rack housing. Figure 1 of this specification shows the wiring diagram governing the electrical connections necessary to the operation of this bomb rack. The electrical system for the rack shall be tested for proper operation as follows:

a. Cock rack with third finger and without aid of special tool and attach electrical connector. Bomb indicator lamp should light.

b. Operate release test switch. Rack hooks shall snap open and bomb indicator light shall turn off and retractive circuit test light shall light.

c. Operate release test switch again. The transfer circuit test lamp should light while switch is held "on".

d. Operate nose arming and chemical test switch. Nose arming solenoid plunger should extend into slot of sliding member and chemical release terminals test lamp should light while switch is held "on".

e. Operate tail arming switch. Tail arming solenoid plunger shall extend into slot of sliding member.

f. Cock rack, operate jettison switch. Bomb rack hooks shall snap open and jettison test lamp shall light as soon as hooks open.

g. Repeat the previous six test procedures for a recheck.

4.4.1.3.1 Each release solenoid shall be energized for a period of 30 seconds at 28 volts dc without resulting damage.

4.4.1.3.2 Each rack solenoid shall have current drain checked as follows:

Solenoid	Amp (Max) 28V dc
Release	6
Tail (Arming)	0.5
Nose (Arming)	0.5

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4.4.1.4 Time of release tests. Measurements shall be made to determine the length of time necessary from beginning of an electrical impulse to the time when bomb carrying hooks begin to open. Not more than .040 seconds shall elapse. Measurement for this test and for that described in 4.4.1.6 are to be made with an 18 volt dc source.

4.4.1.5 Time of transfer tests. Tests shall be accomplished to determine time of transfer. During operation of release circuit the time from end of electrical impulse to completion of switching into next rack circuit shall be a maximum of 17 milliseconds.

4.4.1.6 Each assembled rack shall be inspected for correct solenoid latching and creep resistance using the following procedure:

a. Using a rapid motion, cock the rack with the manual cocking strap. Using the fingers, rotate the solenoid plate counterclockwise. If the movement is less than 2°, the rack is acceptable for this test.

b. Using the fingers, rotate the solenoid plate to the tip of the locking arm. Release the fingers and allow the solenoid to assume its natural position by virtue of the solenoid spring tension. Rotate the solenoid counterclockwise. If the movement is less than 2°, the rack is acceptable for this test.

4.4.1.7 Proof load. The rack shall be proof-loaded after complete assembly by applying a load of 11,000 pounds suspended from the carrying hooks as shown in Figure 2. The position of the loads and cross-section of the pin engaging the carrying hooks shall be in accordance with the dimensions specified on Figure 2. After release, all parts shall function freely.

4.4.1.8 Critical inspection of items:

4.4.1.8.1 Wire cable, leading from the rubber conduit into the solenoid and switch housing, can in no way, under any condition, interfere with the operation of the arming device.

4.4.1.8.2 The cable clamp adjacent to the swivel joint and all parts of the cable swivel unit shall be inspected to see that no burrs or sharp edges will in time sever the insulation of the individual wires.

4.4.1.8.3 The 0.406 $\pm$ 0.001 hole in the brass bushing of the solenoid and switch housing assembly, through which the release lever rotates, is concentric with the 0.281 $\pm$ 0.001 hole in the frame assembly.

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4.4.1.8.4 There is a minimum of 80 per cent bearing between the link lock and tumbler surfaces, and the surfaces lie flat against each other.

4.4.1.8.5 Hooks shall rotate freely after mounting bolts are tight. The inspection shall include the critical dimensions of the associated parts including the dimensions of the length of the shoulders of the pivot bushings, the dimension across the faces of the hook bosses and the size of the frame holes.

4.4.1.8.6 The release lever locking arms shall be checked to insure that the angle and the dimensions which locate the 1/8-inch land with respect to the center of rotation are within the specified limits.

4.4.1.8.7 Binding shall not exist between the pivot hole of the locking lever and the release spring bushing after a short period of operation.

4.4.1.8.8 The 3/16-inch hook fillet, 1/8-inch lock fillet, and 7/64-inch tumbler fillet shall be inspected before plating for freedom from surface irregularities or discontinuities.

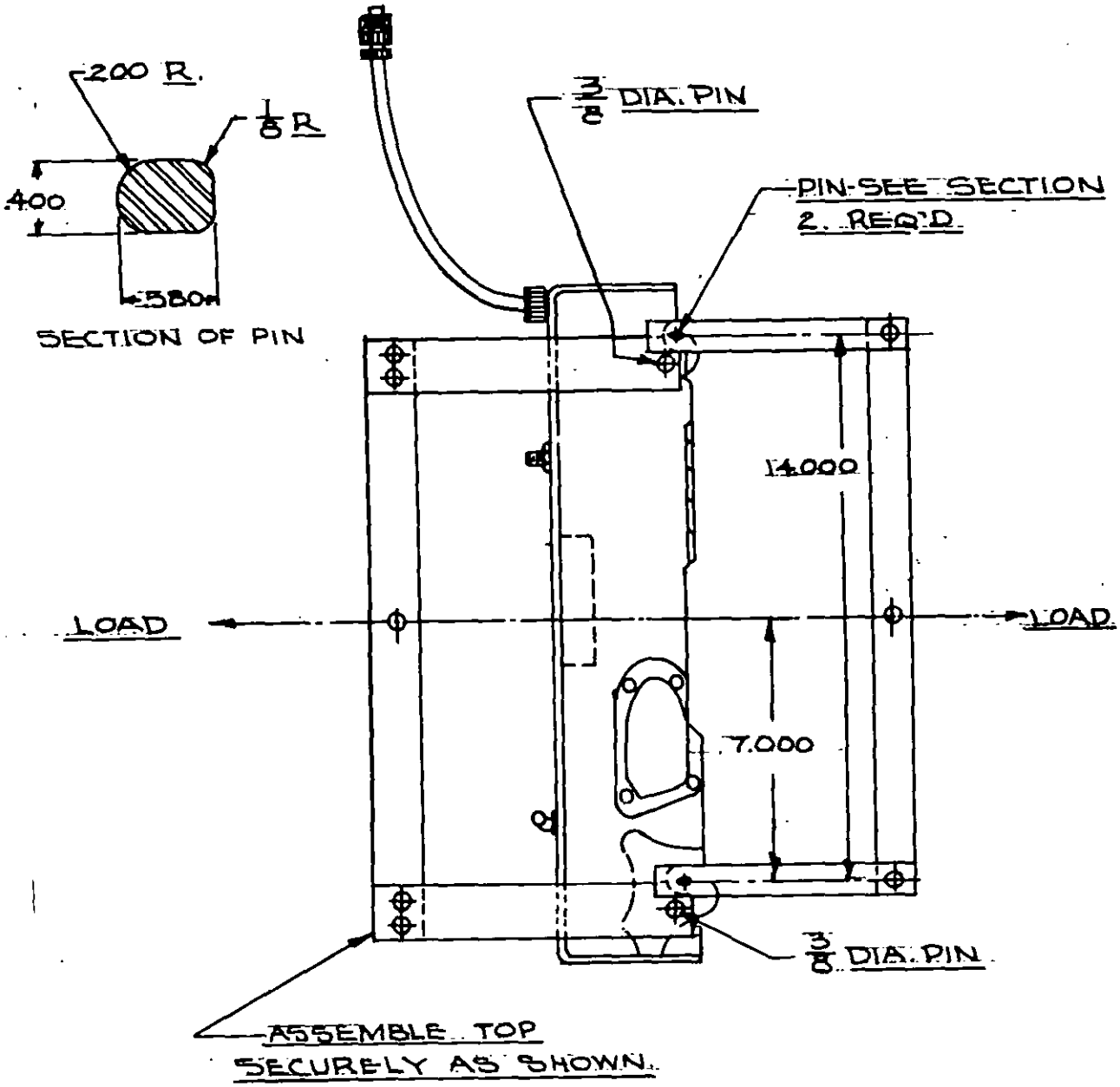
4.4.2 Sampling tests. One rack shall be selected at random from the first lot of 100 or fraction thereof on the order and subjected to the sampling tests specified herein. Thereafter, one rack shall be selected at random from each lot of 500 or fraction thereof and subjected to the sampling tests. These tests shall be in addition to the individual tests specified herein. All racks used in the sampling tests shall be discarded. Test equipment involved shall be subject to the approval of the procuring activity.

4.4.2.1 Lot. A lot shall consist of the Type MA-4A bomb racks made under essentially the same conditions and submitted for inspection at about the same time.

4.4.2.2 Low temperature release and transfer tests. The rack shall be subjected to a temperature of -65°F, or colder, for a period of 2 hours. It shall then be tested at this temperature from an 18 volt dc source. The speed of release shall be as specified in 4.4.1.4. The time of transfer shall be as specified in 4.4.1.5. The arming mechanism shall function satisfactorily as specified in 4.4.1.2.

4.4.2.2.1 After being tested as specified in 4.4.2.2, the rack shall be withdrawn from the cold and allowed to warm up until moisture has collected over the entire rack. The rack shall then be loaded with a maximum of 20 pounds per carrying hook and

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LOADS
PROOF LOAD-11000*
ELEC. RELEASE LOAD
30,000
ULTIMATE LOAD
38000*

TEST FIXTURE

FIGURE 2

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subjected to a temperature of -65°F, or colder, for 2 hours and operated. The speed of release need not be checked; however, the rack shall trip without hesitation and the arming wire retainers must function properly.

4.4.2.2.2 The tests, as specified in 4.4.2.2, shall be repeated at a temperature of 160°F or higher. The rack shall trip and the arming device shall perform correctly.

4.4.2.3 Life test. The rack shall be electrically operated with 10,000 pounds suspended from the hooks through a total of 1000 cycles at the rate of 60 cycles per hour or faster. Each cycle shall consist of cocking the unit and tripping it electrically. At the end of the 1000-cycle operation, no parts shall be broken or damaged in any way as to prevent operation. The first half of the test shall be accomplished using the normal release circuit "B". The remaining portion of the test shall be accomplished using the jettison circuit with sufficient time between cycles to energize a release solenoid connected to the jettison output lead of connector (see Figure 1). At the end of this test the rack shall be checked for a release through the normal release circuit. This test is to be conducted using 28 volts dc.

4.4.2.4 A rack selected at random shall be loaded at 30,000 pounds and released electrically with 18 volts. The rack must release load although damage may result.

4.4.2.4.1 Ultimate strength. The bomb rack selected above shall then be loaded to 38,000 pounds ultimate load and shall be capable of withstanding this load for not less than 10 seconds. The rack shall be mounted using two NAS high shear bolts or equivalent.

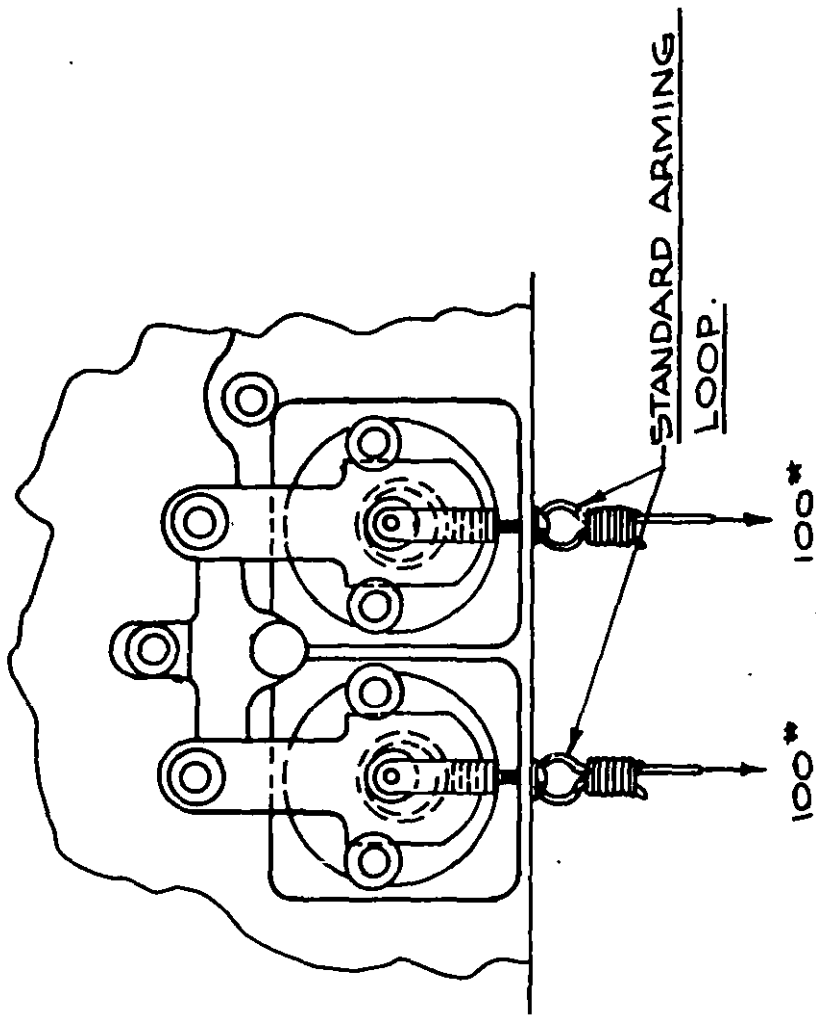
4.4.2.5 Arming wire load. Both of the arming wire retainers shall be able to withstand without failure a downward vertical pull of a minimum of 100 pounds with the rack in the cocked condition (see Figure 3).

4.4.2.6 Arming solenoids life test. The arming solenoids shall be life tested by operation at 28 volts dc for 24 hours. No damage shall result from this continuous operation.

4.4.2.7 Release solenoid test. The release solenoid shall be operated at 28 volts dc for a minimum of 5 minutes. At the end of this test the solenoid shall be operative although damage to insulation or changes in current and voltage characteristics may have resulted.

4.4.2.8 Bomb carrying hooks. Heat treatment of the hooks shall be in lots of not less than 100 and not more than 200 with a minimum

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TEST METHOD, ARM WIRE RETAINERS

FIGURE 3

GPO 826462-2

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Clearance between each hook of 1/2 inch to allow even heat distribution. Also, lot number and date shall be permanently marked on each hook with characters at least 3/32 high with indelible ink.

4.4.2.8.1 A check of the heat treatment of the hooks shall be made by means of test bars heat treated with each lot of hooks. The tensile strength designated on Drawing Nos. 60C98799 and 60C98800 is the minimum allowable and the elongation shall not be more than that specified in the material specification of the grade of steel specified on the drawings. No less than three test bars shall be used with each lot of hooks heat treated.

4.4.2.8.2 One hook shall be tested to destruction from each lot heat treated. The hooks shall be tested in a fixture which simulates their actual assembled position in a cooled bomb rack. Each hook shall withstand a minimum load of 22,000 pounds. One hundred per cent hardness testing shall be conducted for compliance with the drawings.

4.4.3 Rejection and retest. When sampling tests are specified on a quantity of racks that are selected as representative of a certain lot, and one or more of this number fails to meet the specified test(s), acceptance of all items shall be withheld until the extent and cause of failure is determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure, but the final acceptance of the product is contingent upon the Inspectors decision regarding the over-all compliance of the product with specification requirements. If investigation indicates that the defect(s) exist(s) in items previously accepted, full particulars concerning the defect(s) found, including recommendations for correction, shall be furnished to the procuring activity.

4.4.4 Preservation, packaging, packing, and marking. The Inspector shall ascertain that the requirements of section 5 are complied with.

5. PREPARATION FOR DELIVERY

5.1 Unless otherwise specified, preservation, packaging, packing, marking, and testing shall be in accordance with specification MIL-E-17555.

5.1.1 Levels of preservation, packaging, and packing. Unless otherwise specified, preservation, and packaging shall be Level A; packing shall be Level B. Preservation for Level A shall be Method IIb. Unless otherwise specified, the quantity per unit package shall be one each for all levels of packaging.

6. NOTES

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6.1 Use. The Type MA-4A rack as covered by this specification is intended for use in fighter, support and light bombardment aircraft, carrying bombs up to and including 1600 pounds with provisions for carrying and releasing chemicals.

6.2 Ordering data. Invitation for bids, contracts and purchase orders should state the condition for the following:

6.2.1 Preproduction tests. It is expected that the contract or purchase order will specify that one rack will be required as preproduction samples and that these preproduction samples will be subjected to the Preproduction Tests to determine compliance with the requirements of this specification. The invitation for bids and the contract should specify the point of inspection for these tests.

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Preparing Activity
WRAMA

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4009, WHICH IS OBSOLETE C-0270

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